

Leader arm (RB-Y1)

Dynamixel, U2D2, 2 Mbps

joint angles, clutch, trigger

(USB)

PC — teleoperation (Python 3.10)

a) Teleop loop (50 Hz)

FK → cylindrical mapping → command shaping

b) Gripper thread

MX-106, U2D2

c) Camera process

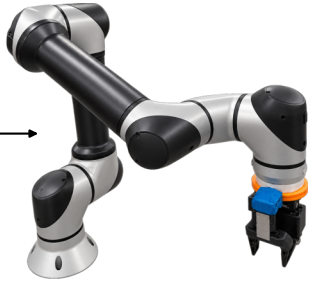
JPEG, 30 fps, one clock

move_servo_l (x, y, z, rx, ry, rz), port 5000

robot state, port 5001

RB5 control box

rbpodo; IK + motor control



RB5-850 with gripper

robot state

move_servo_l

images

Session logs (raw)

states 50 Hz, targets, JPEG 30 fps, one clock

Cameras

front D435, top C270 (fixed); grip Brio 101 (on gripper)

images

LeRobot 0.6.1 (Python 3.12)

Converter

30 Hz grid, nearest frames, filtering

LeRobot dataset

state: 6 joints + gripper
action: x, y, z, gripper
3 cameras 320×240

ACT training

chunk 100, ResNet-18
GPU: RTX A1000

trained policy

Policy runner

Policy process

ACT inference, time-stamped chunks

Servo process (50 Hz)

temporal ensembling 30→50 Hz, same command shaping

data / command flow

dataset flow

autonomous execution